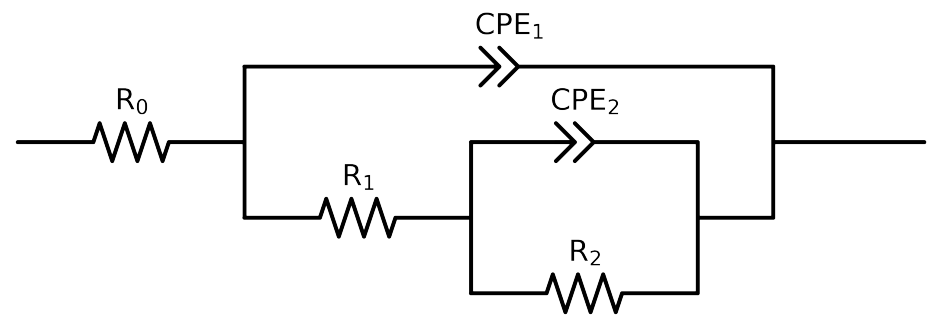


Comparative EIS Kinetics Report

Model Structure: $R_0-p(R_1-p(R_2,CPE_2),CPE_1)$

Used data: original data



Element	Unit	Bounds	PNA_3_1_0h-1	PNA_3_1_1h
R_0	Ω	$[1.0e+00, 1.0e+03]$	$7.84e+01 \pm 2.6e-01$	$9.33e+01 \pm 5.0e-01$
R_1	Ω	$[1.0e+00, 1.0e+08]$	$8.56e+04 \pm 1.3e+03$	$9.98e+04 \pm 2.3e+03$
R_2	Ω	$[1.0e+00, 1.0e+20]$	$1.33e+08 \pm 2.0e-03$	$7.50e+06 \pm 9.4e+00$
Q_2	$\Omega^{-1} \text{ sec}^a$	$[1.0e-09, 1.0e-02]$	$9.64e-07 \pm 6.9e-09$	$8.93e-07 \pm 1.0e-08$
n_2	-	$[5.0e-01, 1.0e+00]$	$8.09e-01 \pm 2.8e-03$	$8.71e-01 \pm 3.6e-03$
Q_1	$\Omega^{-1} \text{ sec}^a$	$[1.0e-09, 1.0e-03]$	$7.95e-07 \pm 5.1e-09$	$8.72e-07 \pm 9.1e-09$
n_1	-	$[5.0e-01, 1.0e+00]$	$8.28e-01 \pm 7.3e-04$	$8.17e-01 \pm 1.2e-03$
χ^2	-	-	$5.574e-05$	$1.648e-04$

Analysis Results: PNA_3.1_0h-1

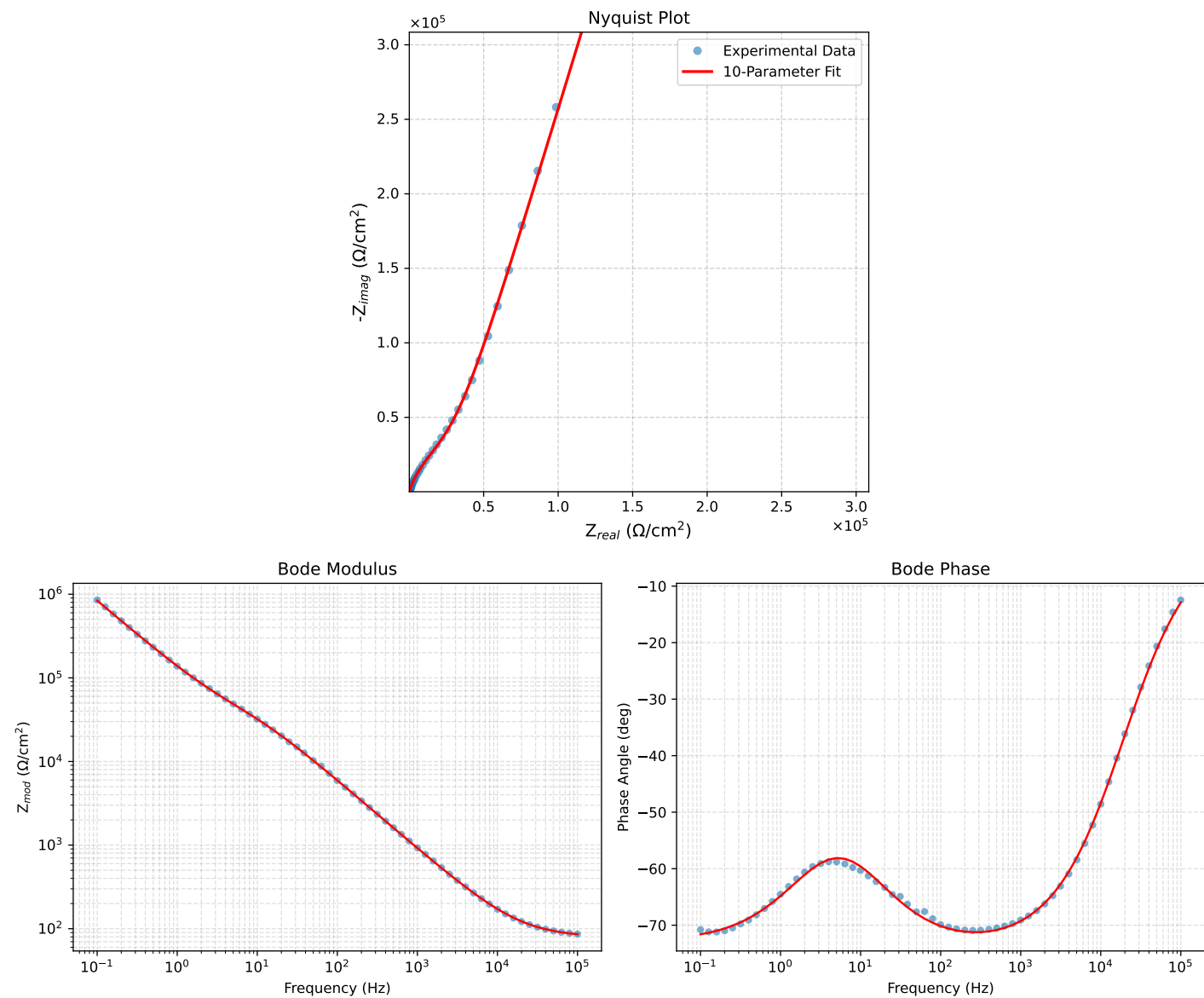


Figure 1: EIS Fit Results for PNA_3.1_0h-1

Fitted Data: PNA_3_1_0h-1

Frequency (Hz)	$Z_{real} (\Omega)$	$-Z_{imag} (\Omega)$	$Z_{mod} (\Omega)$	$Z_{phase} (^{\circ})$
1.0002e+05	8.3742e+01	1.9150e+01	8.5903e+01	-12.8809
7.9512e+04	8.4853e+01	2.3157e+01	8.7956e+01	-15.2646
6.3105e+04	8.6208e+01	2.8040e+01	9.0653e+01	-18.0175
5.0215e+04	8.7829e+01	3.3879e+01	9.4137e+01	-21.0937
3.9902e+04	8.9801e+01	4.0981e+01	9.8710e+01	-24.5297
3.1699e+04	9.2192e+01	4.9582e+01	1.0468e+02	-28.2720
2.5137e+04	9.5112e+01	6.0078e+01	1.1250e+02	-32.2789
1.9980e+04	9.8613e+01	7.2651e+01	1.2249e+02	-36.3802
1.5879e+04	1.0286e+02	8.7868e+01	1.3528e+02	-40.5066
1.2598e+04	1.0804e+02	1.0642e+02	1.5165e+02	-44.5681
1.0020e+04	1.1425e+02	1.2862e+02	1.7203e+02	-48.3867
7.9282e+03	1.2195e+02	1.5611e+02	1.9810e+02	-52.0021
6.3079e+03	1.3109e+02	1.8860e+02	2.2969e+02	-55.1984
5.0347e+03	1.4199e+02	2.2726e+02	2.6797e+02	-58.0023
3.9931e+03	1.5558e+02	2.7526e+02	3.1619e+02	-60.5237
3.1441e+03	1.7269e+02	3.3538e+02	3.7723e+02	-62.7560
2.5195e+03	1.9195e+02	4.0270e+02	4.4610e+02	-64.5151
2.0008e+03	2.1626e+02	4.8713e+02	5.3298e+02	-66.0610
1.5807e+03	2.4663e+02	5.9169e+02	6.4103e+02	-67.3729
1.2536e+03	2.8317e+02	7.1629e+02	7.7023e+02	-68.4300
1.0007e+03	3.2650e+02	8.6237e+02	9.2211e+02	-69.2628
7.9003e+02	3.8219e+02	1.0475e+03	1.1150e+03	-69.9548
6.2881e+02	4.4825e+02	1.2635e+03	1.3406e+03	-70.4664
5.0223e+02	5.2797e+02	1.5191e+03	1.6083e+03	-70.8353
4.0015e+02	6.2693e+02	1.8295e+03	1.9339e+03	-71.0839
3.1550e+02	7.5520e+02	2.2208e+03	2.3457e+03	-71.2191
2.5202e+02	9.0570e+02	2.6658e+03	2.8154e+03	-71.2345
2.0032e+02	1.0965e+03	3.2099e+03	3.3920e+03	-71.1393
1.5801e+02	1.3436e+03	3.8849e+03	4.1107e+03	-70.9220
1.2556e+02	1.6444e+03	4.6676e+03	4.9488e+03	-70.5925
9.9734e+01	2.0229e+03	5.6001e+03	5.9543e+03	-70.1394
7.9449e+01	2.4925e+03	6.6896e+03	7.1388e+03	-69.5645
6.2919e+01	3.0994e+03	8.0058e+03	8.5848e+03	-68.8363
4.9867e+01	3.8620e+03	9.5426e+03	1.0294e+04	-67.9664
3.8422e+01	4.9502e+03	1.1561e+04	1.2576e+04	-66.8202
3.1250e+01	6.0238e+03	1.3399e+04	1.4691e+04	-65.7924
2.4934e+01	7.4461e+03	1.5659e+04	1.7340e+04	-64.5688
2.0032e+01	9.1010e+03	1.8109e+04	2.0267e+04	-63.3172
1.5625e+01	1.1330e+04	2.1205e+04	2.4042e+04	-61.8839
1.2467e+01	1.3678e+04	2.4325e+04	2.7907e+04	-60.6505
9.9734e+00	1.6279e+04	2.7732e+04	3.2157e+04	-59.5862
7.9719e+00	1.9135e+04	3.1546e+04	3.6896e+04	-58.7607
6.3516e+00	2.2242e+04	3.5942e+04	4.2268e+04	-58.2490
5.0134e+00	2.5674e+04	4.1275e+04	4.8609e+04	-58.1175
3.9860e+00	2.9190e+04	4.7428e+04	5.5691e+04	-58.3896
3.1758e+00	3.2893e+04	5.4771e+04	6.3889e+04	-59.0129
2.5161e+00	3.6991e+04	6.3932e+04	7.3862e+04	-59.9466
1.9955e+00	4.1489e+04	7.5100e+04	8.5798e+04	-61.0818
1.5879e+00	4.6476e+04	8.8563e+04	1.0002e+05	-62.3105
1.2581e+00	5.2303e+04	1.0533e+05	1.1760e+05	-63.5922
9.9777e-01	5.9064e+04	1.2571e+05	1.3889e+05	-64.8335
7.9274e-01	6.6966e+04	1.5032e+05	1.6456e+05	-65.9871
6.2903e-01	7.6406e+04	1.8038e+05	1.9589e+05	-67.0427
4.9888e-01	8.7717e+04	2.1692e+05	2.3399e+05	-67.9831
3.9860e-01	1.0083e+05	2.5968e+05	2.7857e+05	-68.7797
3.1672e-01	1.1695e+05	3.1253e+05	3.3369e+05	-69.4836
2.5148e-01	1.3645e+05	3.7661e+05	4.0056e+05	-70.0832
1.9998e-01	1.5985e+05	4.5346e+05	4.8081e+05	-70.5823
1.5879e-01	1.8832e+05	5.4682e+05	5.7834e+05	-70.9971
1.2601e-01	2.2291e+05	6.5984e+05	6.9647e+05	-71.3339
1.0016e-01	2.6456e+05	7.9520e+05	8.3806e+05	-71.5978

Analysis Results: PNA_3_1_1h

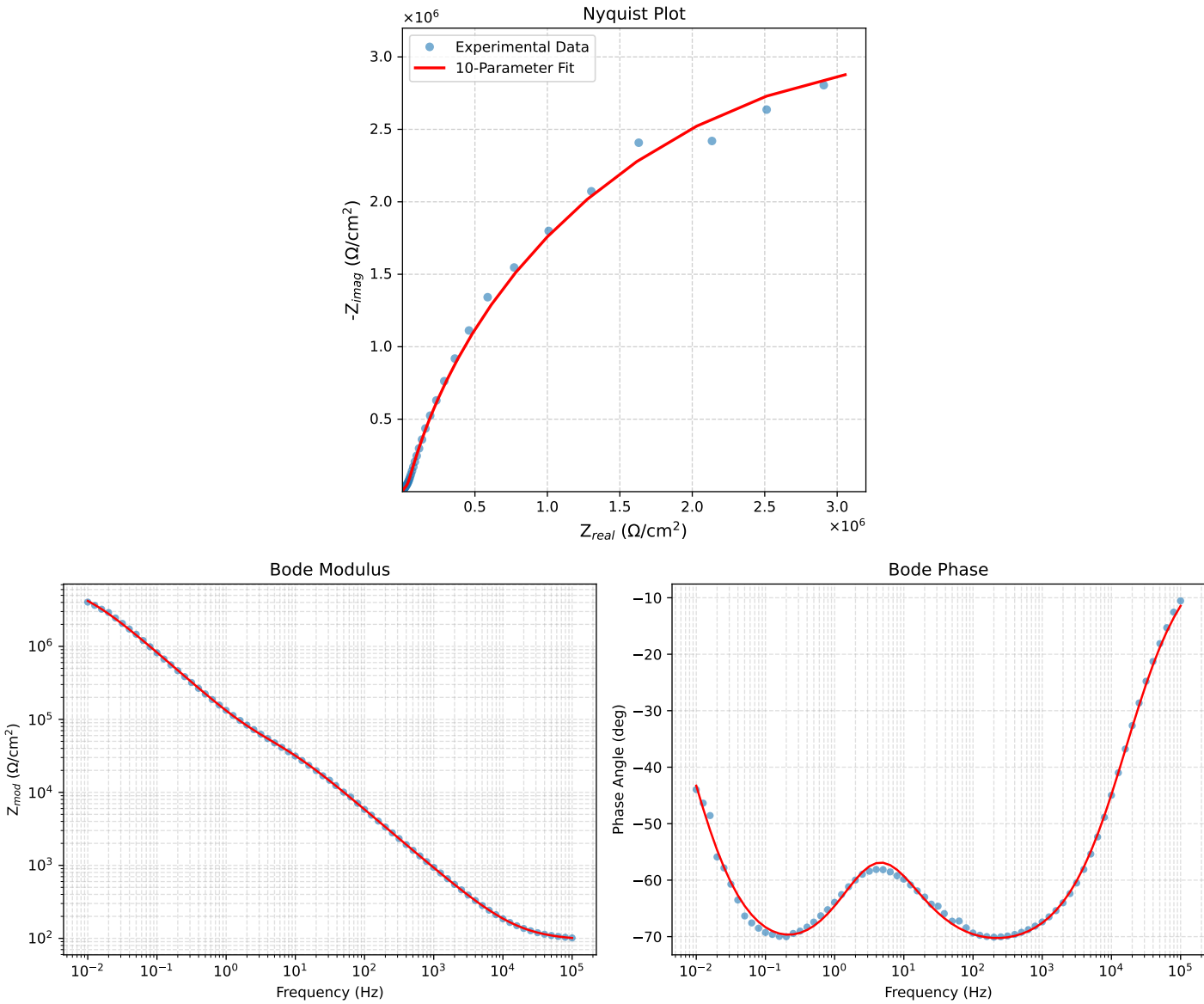


Figure 2: EIS Fit Results for PNA_3_1_1h

Fitted Data: PNA_3_1_1h

Frequency (Hz)	Z_{real} (Ω)	$-Z_{imag}$ (Ω)	Z_{mod} (Ω)	Z_{phase} ($^{\circ}$)
1.0002e+05	9.9169e+01	2.0047e+01	1.0117e+02	-11.4281
7.9512e+04	1.0039e+02	2.4182e+01	1.0326e+02	-13.5431
6.3105e+04	1.0188e+02	2.9208e+01	1.0598e+02	-15.9977
5.0215e+04	1.0365e+02	3.5205e+01	1.0947e+02	-18.7600
3.9902e+04	1.0580e+02	4.2480e+01	1.1401e+02	-21.8754
3.1699e+04	1.0840e+02	5.1269e+01	1.1992e+02	-25.3113
2.5137e+04	1.1157e+02	6.1968e+01	1.2763e+02	-29.0476
1.9980e+04	1.1537e+02	7.4753e+01	1.3747e+02	-32.9418
1.5879e+04	1.1995e+02	9.0188e+01	1.5007e+02	-36.9394
1.2598e+04	1.2553e+02	1.0896e+02	1.6622e+02	-40.9589
1.0020e+04	1.3220e+02	1.3137e+02	1.8637e+02	-44.8201
7.9282e+03	1.4045e+02	1.5905e+02	2.1218e+02	-48.5531
6.3079e+03	1.5020e+02	1.9169e+02	2.4353e+02	-51.9182
5.0347e+03	1.6181e+02	2.3042e+02	2.8156e+02	-54.9220
3.9931e+03	1.7623e+02	2.7841e+02	3.2950e+02	-57.6659
3.1441e+03	1.9432e+02	3.3836e+02	3.9019e+02	-60.1306
2.5195e+03	2.1463e+02	4.0533e+02	4.5864e+02	-62.0983
2.0008e+03	2.4017e+02	4.8913e+02	5.4491e+02	-63.8485
1.5807e+03	2.7194e+02	5.9266e+02	6.5207e+02	-65.3527
1.2536e+03	3.1001e+02	7.1575e+02	7.8001e+02	-66.5815
1.0007e+03	3.5497e+02	8.5974e+02	9.3014e+02	-67.5654
7.9003e+02	4.1247e+02	1.0418e+03	1.1205e+03	-68.4007
6.2881e+02	4.8033e+02	1.2538e+03	1.3426e+03	-69.0380
5.0223e+02	5.6178e+02	1.5042e+03	1.6057e+03	-69.5210
4.0015e+02	6.6231e+02	1.8076e+03	1.9251e+03	-69.8774
3.1550e+02	7.9177e+02	2.1897e+03	2.3284e+03	-70.1201
2.5202e+02	9.4266e+02	2.6233e+03	2.7876e+03	-70.2347
2.0032e+02	1.1327e+03	3.1531e+03	3.3504e+03	-70.2407
1.5801e+02	1.3769e+03	3.8099e+03	4.0511e+03	-70.1302
1.2556e+02	1.6721e+03	4.5712e+03	4.8674e+03	-69.9082
9.9734e+01	2.0410e+03	5.4788e+03	5.8466e+03	-69.5679
7.9449e+01	2.4963e+03	6.5404e+03	7.0006e+03	-69.1096
6.2919e+01	3.0820e+03	7.8261e+03	8.4111e+03	-68.5052
4.9867e+01	3.8162e+03	9.3324e+03	1.0082e+04	-67.7592
3.8422e+01	4.8651e+03	1.1320e+04	1.2322e+04	-66.7438
3.1250e+01	5.9046e+03	1.3140e+04	1.4406e+04	-65.8034
2.4934e+01	7.2931e+03	1.5390e+04	1.7031e+04	-64.6450
2.0032e+01	8.9287e+03	1.7838e+04	1.9948e+04	-63.4103
1.5625e+01	1.1168e+04	2.0934e+04	2.3727e+04	-61.9197
1.2467e+01	1.3571e+04	2.4036e+04	2.7603e+04	-60.5495
9.9734e+00	1.6276e+04	2.7378e+04	3.1850e+04	-59.2691
7.9719e+00	1.9275e+04	3.1043e+04	3.6540e+04	-58.1627
6.3516e+00	2.2541e+04	3.5165e+04	4.1769e+04	-57.3392
5.0134e+00	2.6101e+04	4.0059e+04	4.7812e+04	-56.9135
3.9860e+00	2.9649e+04	4.5640e+04	5.4425e+04	-56.9909
3.1758e+00	3.3251e+04	5.2299e+04	6.1974e+04	-57.5523
2.5161e+00	3.7081e+04	6.0680e+04	7.1113e+04	-58.5710
1.9955e+00	4.1146e+04	7.1037e+04	8.2093e+04	-59.9195
1.5879e+00	4.5564e+04	8.3705e+04	9.5302e+04	-61.4386
1.2581e+00	5.0703e+04	9.9692e+04	1.1184e+05	-63.0421
9.9777e-01	5.6728e+04	1.1935e+05	1.3214e+05	-64.5773
7.9274e-01	6.3925e+04	1.4329e+05	1.5690e+05	-65.9572
6.2903e-01	7.2787e+04	1.7272e+05	1.8743e+05	-67.1488
4.9888e-01	8.3804e+04	2.0866e+05	2.2486e+05	-68.1181
3.9860e-01	9.7113e+04	2.5079e+05	2.6893e+05	-68.8320
3.1672e-01	1.1424e+05	3.0286e+05	3.2368e+05	-69.3331
2.5148e-01	1.3603e+05	3.6583e+05	3.9030e+05	-69.6034
1.9998e-01	1.6362e+05	4.4096e+05	4.7034e+05	-69.6427
1.5879e-01	1.9922e+05	5.3141e+05	5.6753e+05	-69.4498
1.2601e-01	2.4527e+05	6.3947e+05	6.8489e+05	-69.0158
1.0016e-01	3.0446e+05	7.6643e+05	8.2469e+05	-68.3349
7.9503e-02	3.8182e+05	9.1640e+05	9.9276e+05	-67.3807
6.3140e-02	4.8228e+05	1.0902e+06	1.1922e+06	-66.1371
5.0123e-02	6.1318e+05	1.2897e+06	1.4281e+06	-64.5723
3.9833e-02	7.8188e+05	1.5129e+06	1.7030e+06	-62.6691
3.1638e-02	9.9899e+05	1.7577e+06	2.0218e+06	-60.3883
2.5126e-02	1.2743e+06	2.0167e+06	2.3855e+06	-57.7115
1.9957e-02	1.6162e+06	2.2772e+06	2.7924e+06	-54.6347
1.5849e-02	2.0301e+06	2.5221e+06	3.2377e+06	-51.1677
1.2590e-02	2.5134e+06	2.7294e+06	3.7103e+06	-47.3590
1.0001e-02	3.0553e+06	2.8769e+06	4.1966e+06	-43.2780